PH.48111 **FALL 2026**

Autonomous Theory Construction Laboratory

Lecture 10: Validation Sets and Negative Controls

Garam Yun 2026-11-02–2026-11-08

Overview

Negative controls and held-out checks are used to detect leakage, overfitting, and accidental dependence on metadata.

1 Problem definition

The scientific question and the admissible output are specified before any search begins.

$$Barrier = max_{a \in legacy} |R_a| < R_{crit}$$

2 Data and model interface

Data schemas and model interfaces are treated as part of the method, not as post-processing details.

$$\Delta L_{shuffle} \approx 0$$

3 Search and evaluation

Proposal and evaluation are kept separate so that a candidate cannot alter its own test.

$$d^2 = \Delta\theta^T C^{-1} \Delta\theta < d^2_{max}$$

4 Reproducibility

Environment, data version, random seed, and decision record are sufficient to reproduce the reported run.

5 Exercise

The exercise asks for a small implementation and an error analysis rather than a polished narrative.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Yun, ch. 10
- Negative Control Suite NCS-4
- Regression Barrier Guide